TriHop-Bench: Stress-Testing Retrieval on Three- and Four-Hop Questions

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Abstract

We introduce TriHop-Bench, 6,000 English questions requiring three or four reasoning hops, and evaluate three published retrievers using their released code. All of them lose accuracy as hops increase, but the graph-based GW-QA degrades least. The advantage of graph-based retrieval over dense retrieval grows as the number of hops increases.

Keywords: benchmark, multi-hop retrieval, evaluation

1 Introduction

KaveriHop [1] stops at three hops, and older benchmarks such as HotpotQA [5] at two. TriHop-Bench extends evaluation to four hops, where error compounding becomes severe.

2 Benchmark

Questions were generated by chaining three or four encyclopedic passages through bridge entities and then rewritten by annotators. The 3-hop subset has 3,600 questions and the 4-hop subset 2,400. We do not release gold entity links for the test set.

3 Systems

We evaluate DenseHop [2], GW-QA [3] and HopRank [4], each run with the authors' released code and default settings.

Method	Source	TriHop-Bench 3-hop EM	TriHop-Bench 4-hop EM
DenseHop	ours	28.4	9.0
GW-QA	ours	34.7	17.5
HopRank	ours	31.9	11.2

Table 1: Exact match on TriHop-Bench.

4 Results

GW-QA leads DenseHop by 6.3 EM on 3-hop and 8.5 EM on 4-hop questions. The advantage of graph-based retrieval over dense retrieval grows as the number of hops increases.

5 Conclusion

Four-hop questions remain far from solved: the best system reaches only 17.5 EM.

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